Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/03/2005
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Analysis of a Novel Sensory Mechanism in Root Phototropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	10/01/2000	**End Date:**	07/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hangarter, Roger P (Indiana University Bloomington) Edelmann, Richard E (Miami University)		
Grant/Contract No.:	NCC2-1200		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of Arabidopsis. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. The phototropic response in roots is much weaker than the graviresponse, which competes with and often masks the phototropic response. It was through the use of mutant plants with a weakened gravitropism that we were able to identify the activity of the red-light-dependent phototropic system. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. In this project, we plan to use the European Modular Cultivation System (EMCS) on the ISS. We are working toward a launch date of September 2005 on STS-121 for the EMCS and our experiments. This information is important for growing plants on long-term space missions as part of life support systems and advances the current exploration agenda at NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	Past year's Accomplishments--Research/Development Activities 1. ERM-2 Test (Ames Research Center; ARC). This past year, we hired a new post-doctoral associate, Dr. Prem Kumar, to work on the TROPI project. In the July-August 2004 time frame, Dr. Kumar, along with the previous post-doc, Dr. Melanie Correll (now a faculty member at the University of Florida) went to ARC for an EUE (experimental unique equipment) hardware test in the ERM-2 (experiment reference module) of the EMCS (European Modular Cultivation System). This test allowed us to replicate our spaceflight time line. It also gave Dr. Kumar an opportunity to become familiar with the EUE and the procedures at ARC. The PI and Dr. Jack Mullen from Indiana University participated in this trip. We also consulted with ARC staff on final phases of EUE development as well as programming and camera issues regarding EMCS. 2. COSPAR meeting in Paris, France. In July 2005, the PI gave an invited talk on the EMCS EUE at the 2005 COSPAR (Committee on Space Research) meeting. This is an important international meeting for researchers with spaceflight experiments. 3. Functional Test of EUE in FM EMCS at Kennedy Space Center (KSC). In November 2004, the PI and co-PI (Richard Edelmann) assisted in a functional test of the TROPI EUE in the flight model (FM) of the EMCS. At this time, we briefed the astronaut crew (ISS Commander Sergei Krikalev and John Phillips, NASA ISS science officer) of Expedition 11 on our project and the EMCS. The crew indicated enthusiasm for our project after we explained the scientific significance. We also perforemd an abbreviated time line of our experiment to make sure the EUE was successful in the actual FM. 4. Spaceflight and biological procedures development and testing (ARC). The post-doc (Dr. Prem Kumar) was at ARC for two weeks in January 2005 to work with ARC staff on finalizing procedures for seed sterilization and experimental container (EC) assembly. These procedures are critical since they will be used in the preparation for the actual spaceflight experiments. We also tested a new version of the EUE and performed the time line of our space experiment in the ERM-2. 5. Flight build at ARC. This activity is estimated to occur in July 2005. The post-doc and ARC staff will perform the final hardware build under careful QA supervision. The pressure seal on the EUE in EC's (experimental containers) will be tested in the ERM-2. The PI and co-I (Dr. Edelmann) will travel to ARC to approve of the completion of the EUE hardware build. 6. Hardware inspection at KSC. This activity is estimated to occur in July 2005. The PI and co-I will travel to KSC to see if the EUE in EC's were successfully shipped from ARC to KSC. They will throughly inspect the Arabidopsis seeds (biological material) in the EUE to make sure the hardware has met the specified standards. 7. Throughout the funding year. There were numerous telecons among the PI team at Miami University, the co-I at Indiana University, and the ARC team. Detailed minutes from each telecon are available upon request. Some highlights: • Several versions of the EUE have been developed and tested. • Appropriate methods to label seedling cassettes have been developed. • Numerous spaceflight experiment feasibility test have been performed.(e.g. seed viability, long term storage, etc.) • We have continued microarray experiments on root phototropism • Pre-flight and post-flight procedures have been developed. • Numerous videocamera and programing tests of the EMCS have been performed. • We have built up seed stock (with high germination rates) for the spaceflight experiments. At least two generations of seeds will be available of the wild-type and mutant strains of Arabidopsis.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kiss, J.Z.; Correll, M.J.; Edelmann, R.E. "Spaceflight studies of tropisms in the European Modular Cultivation System." International Committee on Space Research (COSPAR) Meeting in Paris, France. COSPAR Abstract Book. , Jul-2004
Abstracts for Journals and Proceedings	Correll, M.J.; Kumar, P. ; Edelmann, R.E.; Bowman, R.N. ; Maldonado, A.; Kakavand, A.; Steele, M.; Eodice, M.; Kiss, J.Z.. "Ground-based studies for a project in development for the International Space Station to study the interactions of gravity and light on plant form." American Society for Gravitational and Space Biology, New York, NY. Gravitational and Space Biology Bulletin 18:53. , Nov-2004

Abstracts for Journals and Proceedings	Molas, M.L.; Correll, M.J.; Kiss, J.Z. "Red-light-induced signaling pathways in Arabidopsis roots." American Society for Gravitational and Space Biology, New York, NY. Gravitational and Space Biology Bulletin 18:23. , Nov-2004
Abstracts for Journals and Proceedings	Molas, M.L.; Correll, M.J.; Kiss, J.Z. "Red-light-induced signaling pathways in Arabidopsis roots." American Society for Gravitational and Space Biology, New York, NY. Gravitational and Space Biology Bulletin 18:23. , Nov-2004
Articles in Peer-reviewed Journals	Correll, M.J.;. Kiss J.Z. 46: 317-323. "The roles of phytochromes in elongation and gravitropism of roots." Plant and Cell Physiology , Feb-2005
Articles in Peer-reviewed Journals	Correll, M.J; Edelmann. R.E.; Hangarter, R.P.; Mullen, J..L.; Kiss, J.Z.. "Ground-based studies of tropisms in hardware developed for the European Modular Cultivation System (EMCS)." Advances in Space Research 35: in press. , Nov-2005
Articles in Peer-reviewed Journals	Lucas, S.Y; Kiss, J.Z. "Roots of Nicotiana sylvestris lack phototropism. " International Journal of Plant Sciences 165: 565-570. , Jul-2004
Articles in Peer-reviewed Journals	Whippo, C,W.; Hangarter, R.P. 2004. "Phytochrome modulation of blue-light-induced phototropism." Plant Cell and Environment , Oct-2004